

Task 3-Hotel Reservation System

Source Code

```
import java.util.*;

public class HotelReservationSystem {
    static ArrayList<Room> rooms = new ArrayList<>();
    static ArrayList<Reservation> reservations = new ArrayList<>();
    static Scanner scanner = new Scanner(System.in);
    static double userBalance = 10000.0;

    public static void main(String[] args) {
        initializeRooms();

        boolean running = true;
        while (running) {
            System.out.println("\n--- Hotel Reservation System ---");
            System.out.println("1. View Available Rooms");
            System.out.println("2. Make Reservation");
            System.out.println("3. View Booking Details");
            System.out.println("4. Exit");
            System.out.print("Choose an option: ");

            int choice = scanner.nextInt();
            scanner.nextLine(); // consume newline

            switch (choice) {
                case 1 -> viewAvailableRooms();
                case 2 -> makeReservation();
                case 3 -> viewBookings();
                case 4 -> {
                    running = false;
                    System.out.println("Thank you for using the system!");
                }
                default -> System.out.println("Invalid option.");
            }
        }

        scanner.close();
    }

    static void initializeRooms() {
        rooms.add(new Room(101, "Standard", 3000.0));
        rooms.add(new Room(102, "Standard", 3000.0));
        rooms.add(new Room(201, "Deluxe", 5000.0));
        rooms.add(new Room(202, "Deluxe", 5000.0));
    }
}
```

```

static void viewAvailableRooms() {
    System.out.println("\n--- Available Rooms ---");
    boolean anyAvailable = false;
    for (Room room : rooms) {
        if (!room.isBooked) {
            System.out.println("Room " + room.roomNumber + " - " + room.category + " - ₹" +
room.price);
            anyAvailable = true;
        }
    }
    if (!anyAvailable) {
        System.out.println("No rooms available at the moment.");
    }
}

static void makeReservation() {
    System.out.print("Enter your name: ");
    String name = scanner.nextLine();

    viewAvailableRooms();
    System.out.print("Enter room number to book: ");
    int selectedRoomNumber = scanner.nextInt();
    scanner.nextLine(); // consume newline

    Room selectedRoom = null;
    for (Room room : rooms) {
        if (room.roomNumber == selectedRoomNumber && !room.isBooked) {
            selectedRoom = room;
            break;
        }
    }

    if (selectedRoom == null) {
        System.out.println("Room not available.");
        return;
    }

    if (userBalance >= selectedRoom.price) {
        selectedRoom.isBooked = true;
        userBalance -= selectedRoom.price;
        reservations.add(new Reservation(name, selectedRoom.roomNumber,
selectedRoom.price));
        System.out.println("Reservation successful. ₹" + selectedRoom.price + " paid.");
    } else {
        System.out.println("Insufficient balance. You have ₹" + userBalance);
    }
}

```

```

static void viewBookings() {
    System.out.println("\n--- Booking Details ---");
    if (reservations.isEmpty()) {
        System.out.println("No reservations found.");
    } else {
        for (Reservation res : reservations) {
            System.out.println("Name: " + res.guestName + ", Room: " + res.roomNumber + ",
Paid: ₹" + res.amountPaid);
        }
    }
    System.out.println("Remaining Balance: ₹" + userBalance);
}
}

```

```

class Room {
    int roomNumber;
    String category;
    double price;
    boolean isBooked;

    Room(int roomNumber, String category, double price) {
        this.roomNumber = roomNumber;
        this.category = category;
        this.price = price;
        this.isBooked = false;
    }
}

```

```

class Reservation {
    String guestName;
    int roomNumber;
    double amountPaid;

    Reservation(String guestName, int roomNumber, double amountPaid) {
        this.guestName = guestName;
        this.roomNumber = roomNumber;
        this.amountPaid = amountPaid;
    }
}

```